
Product Strategy Case Study

Digito

Teaching Mathematics the way good teachers do

Segment	6-8 th Grade (11 – 14)
Strategic Focus	Spaced repetition, adaptive difficulty, peer collaboration
Model	Freemium (Individual) · B2B2C District Licensing
Time Horizon	12-month roadmap, MVP through phase 2 scale

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Business Analysis Portfolio Piece

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Executive Summary

Digito is an adaptive mathematics platform for students in grades 6 through 8 who are falling behind grade-level proficiency. It combines three evidence-based learning science principles - spaced repetition, zone-of-proximal-development scaffolding, and growth-mindset reinforcement - with a feature the category has largely ignored: **structured peer collaboration**, delivered alongside real-time visibility for the classroom teacher.

Where Khan Academy offers breadth without personalization, and IXL offers adaptive drilling without connection, Digito is built on a single insight drawn from twenty-four teacher interviews: students who are behind in math are not failing for lack of content. They are failing for lack of a safe way to ask for help. The product strategy that follows treats that insight as the design constraint it should be.

The go-to-market model is deliberately unglamorous: a six-month MVP validated through teacher-led pilots in three to ten schools, followed by a twelve-month scale phase targeting twenty to thirty districts. Success is measured not by average proficiency gains alone, but by whether the achievement gap between the highest- and lowest-performing quartiles narrows - the metric that actually reflects Digito's mission.

28%	of U.S. eighth graders performed at or above the NAEP Proficient level in math in 2024.
50%	more likely to fail Algebra I when behind grade level entering 6th grade – NWEA
\$3	target price per student, per month — roughly one-tenth the cost of incumbent IXL

Business Context and Problem Statement

Only 28% of eight graders in the United States score proficient or above in mathematics, according to the 2024 National Assessment of Educational Progress. These are not abstract statistics. They describe a specific, compounding failure — students who fall behind in fractions and proportional reasoning in sixth grade rarely catch up, because the middle-school curriculum is sequential and unforgiving.

For entering ninth grade and higher, you need a solid grasp on fundamental mathematics. A student who enters without strong foundation is 2.7 times more likely to fail Algebra I - the single strongest predictor of on-time high school graduation and eligibility for STEM-track coursework. The problem, in other words, is not a testing problem. It is a foundation problem, and it is most severe for exactly the students with the fewest alternative resources: no tutor, no engineer parent to explain fractions at the dinner table, no summer program

Platform	What it does well	Where it fails
Khan Academy	Free, comprehensive, well-produced, beginner-friendly video content.	No personalization to pace or gaps; no teacher visibility into where a student is actually stuck; self-directed structure assumes a confidence struggling learners often lack
IXL	Genuinely adaptive difficulty; widely adopted in schools	Feels like an endless drill with no social or narrative texture; Family (home) membership costs \$79–\$159 per student per year , and for Schools it’s \$30–50 per student annually; no peer or collaborative dimension at all
Private Tutoring	The gold standard for personalized, human interaction	Expensive: \$50–100+ per hour, inaccessible to a large segment of society

The white space is specific: no platform available today combines adaptive, individualized pacing with the social scaffolding and teacher visibility that make struggling students feel safe enough to keep trying. That combination along with affordable pricing - not a smarter algorithm - is Digito's thesis.

Research of Digito

We, at Digito, even before writing a single feature specification, the strategy of Digito was built on direct, structured, science-backed research with three groups who are at the centre of deciding whether an EdTech app survives the challenges that come with the contact with a real classroom: students, teachers, and the administrators who sign the purchase order.

Method	Participants	What we were testing for
Semi-structured interviews	24 middle-school math teachers across 6 districts	Daily workflow, grading burden, trust in existing adaptive tools, willingness to change practice
Contextual student sessions	40 students, grades 6–8, mixed proficiency	Emotional response to being “wrong” in front of peers and teachers; whether makes a hint feel helpful vs. patronizing
Administrator interviews	8 curriculum & instruction directors	Budget cycles, procurement criteria, political risk tolerance for new EdTech, equity mandates
Competitive teardowns	Khan Academy and IXL	Feature parity gaps, pricing benchmarks, onboarding friction, teacher-reported trust

"I don't need another app that tells me Marco is behind. I already know Marco is behind. I need to know why - what specific misconception is holding him back this week."
7th-grade math teacher, middle school, research interview

"I don't raise my hand. Everyone looks at you. I'd rather get it wrong quietly."
— 7th-grade student, research session

These two quotations, more than any market-sizing exercise, define the product: teachers want diagnostic depth without added workload; students want a way to struggle that doesn't feel like exposure or humiliation. Every feature decision in Sections 6 and 7 traces back to one or both of these needs.

Science-Based Learning Methodologies

Digito's product decisions are not based on vague, "intuitive" opinions about what is engaging to students. Each core part of the product relates to a well-replicated, science-backed findings in cognitive and educational psychology.

Principle	Research	Implementation
Spaced retrieval practice	Retrieval at increasing intervals produces more durable learning than re-reading or massed practice (Karpicke & Roediger, 2008; Cepeda et al., 2006)	A scheduling engine resurfaces each mastered concept just as forgetting probability rises
Zone of proximal development	Learning is most efficient in the band between what a student can do alone and what they can do with support (Vygotsky, 1978)	Difficulty adjusts after every response. A struggle triggers a stepped hint - the student completes roughly 80% of the reasoning, the system unlocks only the final step - rather than surfacing the full solution
Growth-mindset framing	Students who attribute ability to effort rather than fixed talent persist longer through failure (Dweck, 2006)	Progress visualizations emphasize effort and improvement trajectory over raw accuracy
Social modeling & self-efficacy	Observing relatable peers succeed at a task measurably increases a learner's belief they can too (Bandura, 1997)	Anonymized peer progress stories surface at moments of friction - "Sofia went from 40% to 75% on this exact skill" - rather than generic encouragement

Taken together, these four principles form a design constraint, not a marketing claim: any proposed feature that cannot be traced to one of them is treated as an assumption to test, not a default to build.

Personas

Three personas in this case study anchor every downstream decision. They are composites drawn from the research above, not idealized archetypes - each carries the specific frictions that killed adoption of prior tools, aim was to maximize the value that wasn't provided by the prior tools.

PERSONA 01 · STUDENT

Sofia

Age & grade	13 · 7th grade
Context	First-generation American, Spanish-speaking household, urban school with 30-student class sections
Where she stands	Roughly 1.5 years below grade level, primarily on fraction and ratio fundamentals from 5th grade
What she wants	To stop feeling like the slow one. To prove - mostly to herself - that she can be good at this
What's in her way	• Embarrassed to ask questions in front of 30 classmates • Math anxiety compounding after years of falling behind • Classroom pace is fixed; no time built in for 1:1 support
What earns her trust	Low-stakes practice, visible incremental progress, and peer voices that sound like her own experience rather than a textbook
Device reality	Primarily a shared family phone; inconsistent home internet

PERSONA 02 · TEACHER

Mr. Chen

Role	7th-grade math teacher, 12 years' experience, 5 sections · 150 students
Weekly reality	4–5 hours grading homework by hand; structurally unable to give each student individualized written feedback
Current tools	Assigns IXL for homework but doesn't fully trust its data - it tells him who is struggling, not why
What he actually needs	• A live view of which concepts the class is collectively stuck on • Diagnostic insight into the misconception, not just the wrong answer • Time returned to him, not new dashboards to babysit
Adoption barrier	Genuine tool fatigue after several EdTech pilots that added work without changing outcomes. Needs a low-friction on-ramp and will abandon anything that feels like extra admin

PERSONA 03 · DISTRICT ADMINISTRATOR

Dr. Martinez

Role	Director of Curriculum & Instruction, mid-size urban district, ~35,000 students
Mandate	Raise district math proficiency with a flat instructional-technology budget
What success looks like to her	• A measurable narrowing of the achievement gap, not just a rising average • No increase in teacher workload or burnout • A defensible, board-presentable case for renewal
How she decides	Evidence of efficacy, total cost of ownership, and political risk - a visible failure with a new vendor carries real career cost, so she moves cautiously and wants pilot data before a district-wide contract

Feature Strategy

6.1 MVP Scope - Month 1 through 6

The MVP is scoped tightly around Sofia and Mr. Chen. Dr. Martinez's district-scale needs are deliberately deferred: her trust is earned through pilot evidence, not built in advance of it.

Feature	Why it belongs to MVP	Scientific Root	Serves
Adaptive problem engine	The load-bearing mechanic;	Spaced retrieval + ZPD	Sofia
Stepped worked examples	Prevents learned helplessness at the moment of struggle	ZPD scaffolding	Sofia
Student progress view	Makes incremental improvement visible and motivating for student	Growth mindset	Sofia
Teacher insight panel	Answers “where is my class actually stuck”	Diagnostic retrieval data	Mr. Chen
Curated async peer help	first version of peer collaboration	Social modelling	Sofia & Mr. Chen
Mobile-first interface	Reflects how Sofia actually accesses the internet	Access & equity	Sofia

6.2 Phase 2 - months 7 through 12

Real-time teacher collaboration. Mr. Chen receives an alert when a student is stuck for more than a threshold period and can join a brief live session - scaffolding the reasoning without solving the problem for the student.

Misconception reporting. Pattern analysis across wrong answers surfaces specific, named misconceptions (for example, “student treats negative numbers as invalid solutions”) so Mr. Chen can plan a five-minute targeted re-teach rather than a full-period review.

Parent visibility (light-touch). Parents see trend-level progress, not assignment-level detail, paired with a one-click explainer - “how to help with fractions this week” - addressing Dr. Martinez's engagement mandate without adding surveillance anxiety.

6.3 What we deliberately did not build

Considered	Decision	Reasoning
Points & leaderboards	Excluded from MVP	Extrinsic reward structures can undermine intrinsic motivation in math specifically (Pink, 2009); Sofia's cohort already carries math anxiety, and public ranking risks amplifying it. Will A/B test an opt-in, effort-based version in Phase 2
Open-ended AI tutor	Deferred, not rejected	Evidence for open-ended AI tutoring with low-engagement learners is mixed; teachers in research consistently wanted to stay in the pedagogical loop; and the cost/complexity is not justified before MVP signal exists

Product Roadmap

Four phases, each gated by evidence rather than a calendar date. A phase does not advance until its success metric is met; the timeline below reflects the planned, not guaranteed, pace.

Phase	Timeline	Core deliverables	Gate to advance
MVP Alpha	Weeks 1-8	Adaptive engine across 5 topics (fractions, decimals, percentages, ratios, exponents); student dashboard; basic teacher insights; mobile UI; closed beta with 3 teachers, 60 students	99% uptime · 20% rise in weekly problems solved · NPS above 40
MVP Refine	Weeks 9-16	UX fixes from beta feedback; hint-engine tuning; async peer collaboration v1; Google Classroom integration; 8 additional topics	70% weekly active use · 80% of problems land in optimal difficulty range
Market Validation	Weeks 17-26	Paid pilot launch across 5–10 schools (500–1,000 students); freemium tier live; \$3/student/month school pricing; independent efficacy read	80% school renewal intent · 15%+ assessment gains vs. control group
Phase 2 Scale	Months 7-12	Live teacher collaboration; misconception reporting; parent portal; grade-band expansion to 5–9	20–30 schools active · 3–5 point narrowing of the achievement gap

Success Metrics

8.1 Student-level indicators

Metric	Month 6 target	Why it matters
Weekly active students	> 70% of enrolled	The floor for any downstream learning claim
Time on platform	> 90 min / week	Proxy for genuine, not compliance-driven, adoption
Accuracy trend	+10 pts, quintile 1 → 3	Directional evidence the adaptive loop is working
Student NPS	> 40	Whether Sofia would tell a friend

8.2 Business indicators

Metric	Month 6	Month 12
Paying school pilots	5–10	20–30
Monthly recurring revenue	\$5K–\$10K	\$30K–\$50K
Monthly logo churn	< 10%	< 5%
Teacher NPS	> 35	> 50

8.3 The metric that overrides the others

The gap between the highest- and lowest-performing quartiles of students must narrow by 3–5 percentage points within the first cohort's first year. If Digito improves outcomes for already-strong students faster than it improves outcomes for students like Sofia, the product has failed its stated purpose - regardless of what the average proficiency number shows.

Go-to-Market Strategy

Our research has determined one belief which is - teachers, not district procurement, are the real gatekeepers of classroom adoption. Trust is earned bottom-up before it is sold top-down. We are preparing the launch sequence built around that tested belief.

Phase	Approach	Why it works
1 • Months 1–4	Partner with 3-5 “innovator” teachers in target schools. Free access to one class section in exchange for weekly feedback and case-study participation from them.	Removes financial risk entirely; builds authentic product feedback loop before any sales motion begins
2 • Months 5–8	Expand to 5-10 schools at a subsidized pilot rate (\$1–2/student/month) bundled with 3 hours of teacher onboarding and monthly office hours	Below-market pricing plus built-in support makes the pilot an easy, low-stakes yes for Dr. Martinez
3 • Month 6+	Open a freemium tier for individual students outside any school relationship; premium features and school-pathway prompts drive conversion	Generates bottom-up demand - a parent or student requesting their school adopt Digito is the cheapest possible acquisition channel

This sequencing is intentional: **teacher trust → pilot evidence → administrator confidence → peer-driven demand**. Reversing the order - selling to a district before any classroom has used the product - is the single most common failure pattern observed across three four competitor teardowns in Competitive Audit Section .

Competitive Audit

Competitor	Strength	Weakness	Digito's answer
Khan Academy	Free; deep, well-produced content library; 50M+ users	No personalization to pace; no teacher visibility; self-directed structure assumes confidence struggling learners lack	Adaptive pacing plus a teacher insight layer better than Khan Academy has built
IXL	Genuinely adaptive; strong existing school penetration	Feels like an endless drill; \$30-50 per student/year; zero peer or social dimension	One-tenth the price with collaboration and diagnostic depth built in from day one
Private tutoring	Human, personalized, trusted	\$50–100+ per hour; structurally inaccessible at the scale of the problem	Scaled, affordable approximation of individualized attention via peer + teacher loop

Why this positioning holds

Evidence over intuition. Every mechanic traces to named research, not a hunch about what feels engaging.

Collaboration as the wedge. Adaptive-plus-peer-plus-teacher is a combination no incumbent has shipped, because it requires trusting teachers with data rather than replacing them.

Price built for the mission. At roughly one-tenth IXL's cost, Digito is affordable precisely for the districts the mission is built around - not despite the equity focus, but because of it.

Risks & Assumptions

11.1 What has to be true, and how we'll know

Assumption	Validation method	Contingency if wrong
Students sustain 90+ min/week of use	Track weekly active usage across the 8-week beta cohort	Shift to a teacher-assigned, classroom-required model rather than a self-directed one
The adaptive engine measurably improves accuracy	Compare pre/post accuracy over a 4-week window	Increase hint scaffolding density; reduce the size of difficulty jumps between steps
Teachers act on dashboard insights	Qualitative interviews plus direct classroom observation	Replace passive dashboards with push notifications and pre-summarized action items
Schools will pay \$3/student/month post-pilot	Willingness-to-pay conversation built into every pilot exit	Test outcome-linked pricing - payment tied to measured proficiency gains

11.2 External risk

Data privacy (FERPA). Mitigated through encrypted storage, no third-party data sharing, and an annual third-party security audit — budgeted at roughly \$50K per year.

Platform risk from foundation-model tutors. A free, general-purpose AI tutor could commoditize content delivery. Mitigation is relational, not technical: teacher relationships and the collaboration mission are harder to displace than content quality alone.

Adoption fatigue. Teachers have been burned by EdTech pilots before. Mitigation is structural: onboarding is designed to save time from week one, not add a new system to maintain.

Conclusion

Digito starts from a problem that is well documented and badly served: middle-school students who fall behind in mathematics rarely catch up, and the tools built to help them are either impersonal at scale (Khan Academy), mechanical and isolating (IXL), or personal but unaffordable (tutoring). None of them solve for the thing research surfaced again and again - students like Sofia don't need more content. They need a way to struggle that doesn't feel like exposure and humiliation.

The strategy in this document is deliberately narrow in its first six months: prove, with three to ten real schools, that adaptive pacing plus structured peer support plus teacher visibility measurably narrows the achievement gap - not just the average score. Everything else, including the district-scale features Dr. Martinez will eventually need, is sequenced to come after that evidence exists, not before it.

If the strategy is right, Digito earns its place not by out-marketing Khan Academy or undercutting IXL on price alone, but by being the first platform whose central metric is the one that actually matters: whether the students furthest behind are catching up.

This case study was prepared as a portfolio piece demonstrating product strategy, learning-science research application, and structured business analysis for entry-level Business Analyst roles.